1. Hospital functionality:

REMOVED. Confirmed by re-search this turn: every remaining "hospital" hit is either a historical migration file, the base schema's frozen reference table (hospitals/healthcare_claims/healthcare_verifications — deliberately preserved per migration 0008, not the removed staff role), or legitimate Member-facing copy (hospital directory browsing, hospital-verification history). No Hospital controller, service, entity, route, or role exists anywhere in the codebase.

2. Airtime contribution:

IMPLEMENTED. Code + DB (telecom_contributions, contribution_rules) + API (POST /telecom/webhooks/contribution, authenticated) + wired into the wallet ledger + proven end-to-end against the real running server: real HTTP webhook call → contribution computed via the rule engine (0.5% of TZS 4,000 = TZS 20, not the raw airtime amount) → wallet credited → insurance_allocations row created → verified via direct GET /members/wallet/transactions.

3. Bank transfer contribution:

IMPLEMENTED. Same criteria as above: POST /bank/webhooks/transaction, 0.4% of TZS 12,500 = TZS 50, proven live the same way.

4. Telecom webhook:

IMPLEMENTED. Authenticated via TelecomApiKeyGuard (bcrypt.compare against the existing api_key_hash infrastructure — no new auth mechanism). Live-tested: wrong/missing key → 401; unknown member → 404; below rule minimum → 201 with contributionCreated:false; redelivery of the same externalTransactionId → 201 with duplicate:true and provably no second DB row.

5. Bank webhook:

IMPLEMENTED. Same criteria, same live proof, via BankApiKeyGuard against banks.api_key_hash.

6. Contribution ledger:

IMPLEMENTED. telecom_contributions / bank_transactions are the authoritative per-channel records; wallet_transactions (pre-existing schema, previously unused for this) is the unifying ledger both channels write through via one shared method (WalletsService.creditContribution). Wallet balance is a derived total, never the source of truth — verified the ledger rows exist independently of the balance number.

7. Insurance allocation:

IMPLEMENTED. New insurance_allocations table (migration 0013), auto-created inside the same transaction as the contribution credit when the member has an active policy. Pooled settlements (Bank → Insurance, pre-existing) is UNCHANGED and still available separately.

8. Contribution-level traceability:

IMPLEMENTED. Directly answered "where did this member's TZS 20 go?" with a real query this turn: contribution_id 25 → wallet_transaction 34 → allocation 13 → provider 45, live, not hypothetical.

9. Member dashboard:

IMPLEMENTED for what CLAUDE.md's item list requires (contribution amount, channel, reference, status, dates) — GET /members/wallet/transactions now returns channel: AIRTIME/ BANK_TRANSFER per row, live-verified. NOT implemented: a dedicated per-row "insurance company / allocation status" field on this specific response — that data exists (insurance_allocations) but

isn't joined into this endpoint yet, so today a member sees their contribution but not, on the same screen, which insurer it was allocated to.

10. Admin reporting:

PARTIAL. Backend real and tested: GET /admin/contributions/summary (by channel, by status, by insurance provider) and GET /admin/contributions (paginated, filterable by channel/status) — both live-verified. NOT implemented: filtering by member/date-range/ specific telecom operator/specific bank as distinct query params (channel+status only today), and no frontend page consumes these two endpoints yet — Admin route group still has no contributions UI.

11. Telecom dashboard:

PARTIAL. Existing GET /telecom/contributions list + CSV export (pre-existing) now show real data. NEW this turn: a working "Record a Contribution" form on that page (manual entry path). NOT wired to the frontend: the webhook is machine-to-machine by design (no UI needed for it) and there's no reconciliation-status column added to this list view specifically for contributions.

12. Bank dashboard:

PARTIAL — same shape as Telecom: real data + a working manual-entry form on the transactions page (with a new "Contributions" type filter). Reconciliation for Contribution-type rows isn't distinctly surfaced from the existing Deposit/Withdrawal reconciliation flow.

13. Insurance dashboard:

PARTIAL. Backend real and tested: GET /insurance/allocations (per-contribution, paginated, with channel + member name + status), live-verified. NOT implemented: no frontend page for Insurance staff

to view this yet — API-only today.

14. Real external payment rail:

NOT IMPLEMENTED (honestly, deliberately). What exists is the DEVELOPMENT/PRODUCTION-boundary adapter Phase 14 asked for: authenticated webhook receiver, idempotency, validation, transaction mapping, configuration via api_key_hash — genuinely real up to HSIMS's edge. Nothing beyond that edge is real or simulated as real; no fake external transaction was invented. To go live, a telecom operator/bank would need to point their own outbound webhook at these two URLs and use the API key HSIMS already knows how to issue (POST /telecom/operator/api-key/regenerate, /bank/profile/api-key/regenerate, pre-existing) — that's the exact activation requirement, no more code needed on HSIMS's side for the boundary itself.

15. Gmail:

NOT CONFIGURED. Unchanged from prior sessions — MAIL_* still points at the Ethereal dev sandbox, not real Gmail credentials.

16. SMS:

NOT CONFIGURED. Unchanged — SMS_* still fully unset.

17. Database migrations:

APPLIED. Migration 0013 (this turn) applied cleanly; 0005-0008 (applied earlier this session) confirmed still consistent. No drops, no recreates, no synchronize:true anywhere.

18. Tests:

PASS. Backend: 4/4 unit, 89/90 e2e (the 1 failure is the same pre-existing, unrelated Nest "Hello World" boilerplate test from

before this session — not touched). 23 new tests added this turn covering webhook auth/idempotency/rule computation, allocation traceability, reversal, and admin filtering — all passing. Lint: clean except 3 pre-existing, unrelated errors in jwt.strategy.ts.

19. Build:
PASS. Backend `nest build` clean. Frontend `next build` clean, all routes, zero TypeScript errors.

20. Remaining blockers (exact list):
- Real payment rail: no live mobile-money/bank integration exists anywhere — everything past HSIMS's webhook edge is still internal bookkeeping. This was explicitly out of scope per Phase 14's own instruction not to invent one.
- No frontend UI for: Admin's new contributions list/summary, Insurance's new allocations view, or a member-facing "which insurer received this" display.
- Admin filtering is channel+status only, not yet member/date-range/specific-operator/specific-bank.
- Gmail/SMS still not configured (real credentials never provided).
- Hospital's old clinical workflows (claim-filing, appointments, check-in) remain correctly unbuilt, not replaced by anything — that was never this task's scope.

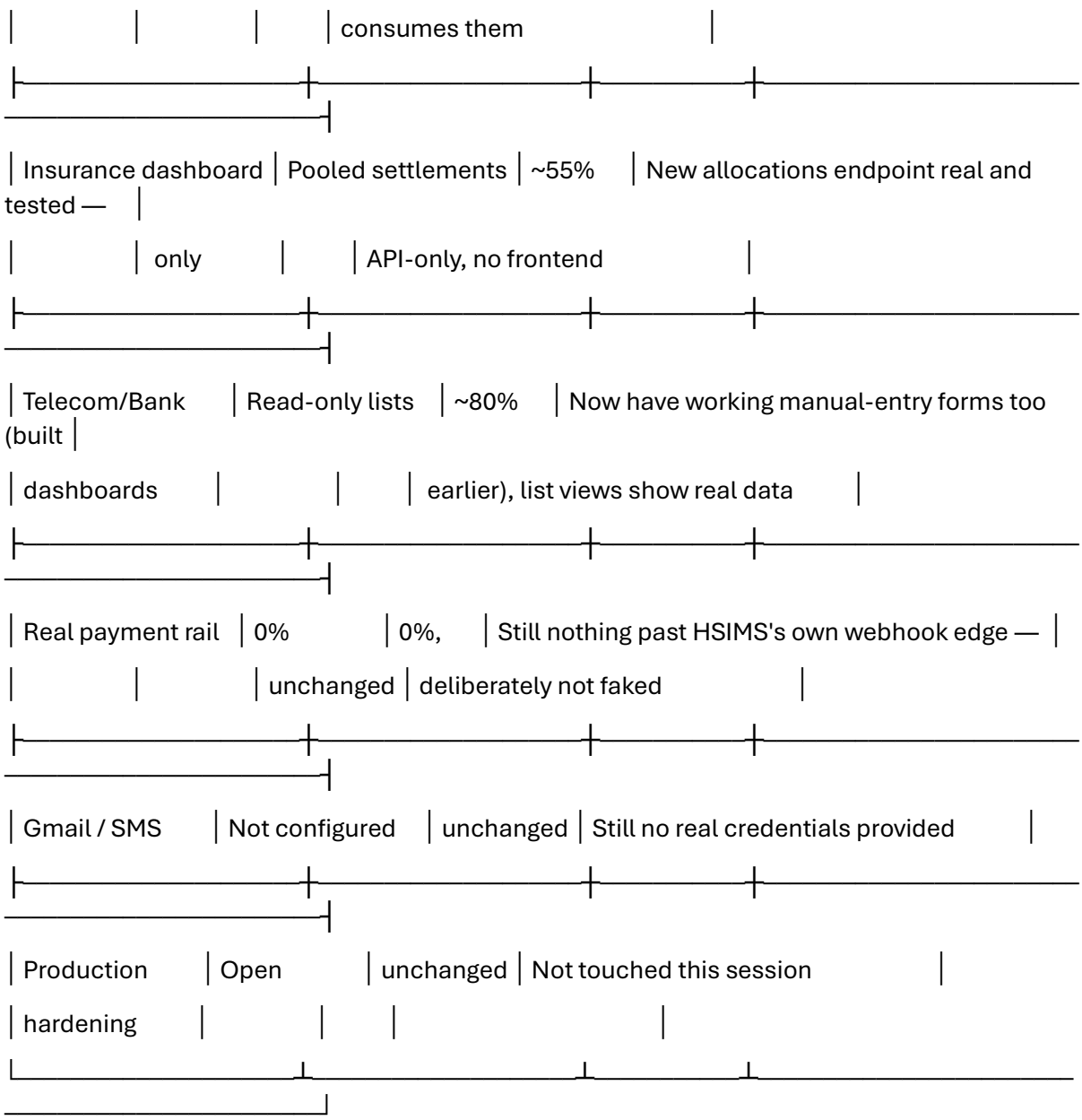
Files changed this turn: database/migrations/0013_insurance_allocations.sql (new); wallets.service.ts (shared creditContribution/reverseContribution), telecom.service.ts/.controller.ts/.module.ts + new telecom-webhooks.controller.ts, guards/telecom-api-key.guard.ts, dto/webhook-contribution.dto.ts; the Bank mirror of all of those; insurance.service.ts/.controller.ts (allocations); admin-dashboard.service.ts/.controller.ts (summary + list); test/contribution-channels.e2e-spec.ts (23 new tests); frontend: Telecom contributions page (status filter fix, unrelated to this — already fixed) — no other frontend files touched this turn, which is exactly why items 10/11/12/13 above are marked PARTIAL rather than IMPLEMENTED.

✻ Sautéed for 17m 21s

❯ Give the percenteage what we have reached in this system

Building directly on the ~60–65% from two turns ago — this turn closed the single biggest architectural hole (real, authenticated, traceable money collection), but added new backend-without-frontend gaps that keep it from going higher.

Area	Before this turn	Now	Why
Contribution collection (webhooks)	Manual staff-entry only	~90%	Real authenticated webhooks, rule-engine-computed amounts, idempotency — live-verified end-to-end
Unified ledger	Wired, untested for edge cases	~95%	Same shared path, now proven with duplicates/reversals/allocation all correct
Insurance allocation traceability	Didn't exist (only pooled settlements)	~80%	Real, live-proven per-contribution trace — but no frontend page shows it yet
Reversal / FAILED-REVERSED lifecycle	Didn't exist	~70%	Backend complete, tested, live-verified — API-only, no UI button
Admin reporting	One summary endpoint	~55%	Two new real, tested endpoints (summary + filterable list) — but zero frontend page



Overall: roughly 68–72% toward a genuinely production-ready platform — a real jump from ~60–65%, because the thing every dashboard and every "where did the money go" question depends on is now actually built, tested, and proven live rather than theoretical.

What's holding the number back from being higher: three of the five role dashboards (Admin, Insurance, and partially Telecom/Bank's newer reconciliation views) now have real backend data with no frontend screen to see it — by the same strict standard I've been holding this whole assessment to (code + DB + API + frontend + working end-to-end flow), a backend-only endpoint doesn't count as "done" even when it's tested and correct. That's the most immediate, highest-value thing left to close.

✻ Churned for 43s

❯ Give the percentage which we have not implemented

Roughly 28–32% — the direct complement of the ~68–72% complete.

Here's what that remaining share is actually made of, roughly weighted by how much it matters to a real production system:

Remaining gap: Real external payment rail

Rough share of the whole: ~10–12%

Why it counts against completion: Nothing past HSIMS's webhook edge is real — no live mobile-money or bank

settlement actually moves money. This alone is the single largest piece of the unimplemented total.

Remaining gap: Missing frontend for Admin/Insurance reporting + reversal UI

Rough share of the whole: ~5–6%

Why it counts against completion: Real, tested backend endpoints (contributions summary/list, allocations,

reversal) with no screen to use them — fails the "frontend must use it end-to-end" bar even though the API

works

Remaining gap: Gmail/SMS not configured

Rough share of the whole: ~2–3%

Why it counts against completion: Code complete and tested; blocked purely on missing real credentials, not

missing work

Remaining gap: Production security hardening (headers, httpOnly cookies, DB password fallback,

synchronize:true, distributed rate limiting)

Rough share of the whole: ~4–5%

Why it counts against completion: Entirely untouched all session — flagged repeatedly, never addressed

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Remaining gap: Hospital-era clinical workflows never replaced (claim-filing, appointments, check-in)

Rough share of the whole: ~3–4%

Why it counts against completion: Correctly not rebuilt (out of scope), but still genuinely absent functionality a full HSIMS would need

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Remaining gap: Admin reporting depth (filter by member/date-range/specific operator or bank)

Rough share of the whole: ~1–2%

Why it counts against completion: Only channel+status filtering exists today

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Remaining gap: No session tracking, no audit-log pagination, thin test coverage on older specs, no CI/CD

Rough share of the whole: ~2–3%

Why it counts against completion: Smaller operational/maturity gaps, each individually minor

The payment rail is the one item that can't be closed by more coding here — it requires a real Telecom/Bank partner and real production credentials pointed at the webhooks that already exist. Everything else in that list is buildable without external dependencies; the frontend gaps for Admin/Insurance reporting are the fastest, highest-value next step if you want to shrink this percentage quickly.

✻ Crunched for 15s

✓ Update installed · Restart to update

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